

I-Hsun Chen

Curriculum Vitae

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Research Interests

Random graphs, large deviations, interacting particle systems, spin glasses.

Education

- 2021–present **Brown University**, Providence, RI
Ph.D. Candidate, Applied Mathematics.
Expected Graduation Date: May 2027.
- 2024 **Brown University**, Providence, RI
ScM in Applied Mathematics.
- 2016–2020 **National Taiwan University**, Taipei, Taiwan
B.S. in Mathematics.

Research Experience

- 2021–present **Ph.D. Student**, Brown University, Providence, RI
Advisor: Kavita Ramanan.
- Summer 2021 **Research Assistant**, National Center for Theoretical Sciences, Taipei, Taiwan
- 2020–2021 **Research Assistant**, Academia Sinica, Taipei, Taiwan
Advisors: Ju-Yi Yen, Chii-Ruey Hwang.
Excursion Approach to Central Limit Theorem for Some Null Recurrent Diffusion Processes.
- 2019–2020 **Undergraduate Research**, National Taiwan University, Taipei, Taiwan
Advisor: Chii-Ruey Hwang.
Accelerating Markov Chain Monte Carlo method through stochastic differential equations driven by Lévy processes.

Publications

1. **I-Hsun Chen**, Ivan Lee, Kavita Ramanan, Sarath Yasodharan. “Gibbs Conditioning, Atypical Consensus and Splitting Gibbs Measures on Random Regular Graphs.” Accepted to *Mathematical Physics, Analysis and Geometry*. (special issue in honor of Claudio Landim's 60th birthday.) arXiv:2603.01322
2. **I-Hsun Chen**, Kavita Ramanan, Sarath Yasodharan. “Large Deviation Principles for Neighborhood Empirical Measures of Contextual Stochastic Block Models.” Preprint, 2026.
3. **I-Hsun Chen**, Kavita Ramanan. “Large Deviation Principles for Colored Configuration Models” In preparation, 2026
4. **I-Hsun Chen**, Kavita Ramanan. “Large Deviations for Marked Erdős-Rényi and Scale-Free Simple Configuration Models” In preparation, 2026

Conferences and Workshops

- Mar 2026 **Seminar on Stochastic Processes**
Lightening talk and Poster: Gibbs conditioning and atypical consensus on random regular graphs
- Dec 2024 **BIRS-CMI Research School: Mathematical Foundations of Network Models and Their Applications**
Student lead of the project: Bethe Prediction for Statistical Physics Models with Non-pairwise Interactions
- Aug 2024 **Princeton Machine Learning Theory Summer School**

May 2023 **Graphical Models, Statistical Inference, and Algorithms**
Jun 2022 **PIMS-CRM Summer School in Probability**

Invited Talks

July 2025 **Large Deviations of the Neighborhood Empirical Measure of Heterogeneous Marked Sparse Random Graphs**, *Mathematical Congress of the Americas 2025*, Miami, FL
June 2025 **A Sanov-type Theorem for Marked Sparse Random Graphs**, *INFORMS APS 2025*, Atlanta, GA
December 2024 **Large Deviations of the Empirical Measure of Heterogeneous Sparse Graphs with Polish Spaced Valued Marks**, *BIRS-CMI workshop "Mathematical foundations of network models and their applications"*, Chennai, India
2023 **A Sanov-type Theorem for Unimodular Marked Random Graphs and Its Applications to Spin Models**, *NTU Math Colloquium*, Taipei, Taiwan
2021 **Brownian Additive Functionals Averaged by Local Time**, *Ritsumeikan University Mathematical Finance Seminar*, Kyoto, Japan. (Virtual)

Awards

2024-2026 **Government Scholarship to Study Abroad**, *Ministry of Education*, Taiwan (R.O.C.)
2019 **Scholarship for Excellence**, *National Taiwan University*
2017-2019 **Sun-Yung A. Chang & Paul C. Yang Scholarship**, *National Taiwan University*
2019 **The 10th S.-T. Yau College Student Mathematics Contests**, *Honorable Mention in Probability and Statistics*
2015 **Asian Pacific Mathematical Olympiad**, *Honorable Mention*

Advising and Mentoring

2025-2026 **Ritvik Teegavarapu**, *California Institute of Technology*, SURF Fellows Program
Gibbs Conditioning Principles for Dynamical Models on Sparse Random Graphs
Co-advisor, with Kavita Ramanan.
2023-2025 **Ivan Lee**, *Brown University*, Undergraduate Research and Honor Thesis
Gibbs Conditioning Principles for Marked Erdős-Rényi and Random Regular Graphs
Co-advisor, with Kavita Ramanan and Sarath Yasodharan.
2023-2024 **Yizhong Hu**, *Brown University*, Undergraduate Research and Honor Thesis
Annealed Pressure of Nearest-Neighbor Non-pairwise Interaction Spin-Glass Models on Random Regular Graphs (J.L. Stein award for undergraduate excellence in applied mathematics)
Co-advisor, with Kavita Ramanan and Sarath Yasodharan.

Service and Teaching

2022 **APMA Directed Reading Program**, *Brown University*
Mentor for "Mean-Field Models in Spin Glasses"
2020 **Summer Student Seminar**, *National Taiwan University*
Co-organizer and Lecturer of the Analysis Section
Teaching Assistant, *Brown University*
Operation Research: Probabilistic Method (Spring 2023), Statistical Inference (Fall 2022)
Teaching Assistant, *National Taiwan University*
Stochastic Calculus
Teaching Assistant, *National Taiwan University*
Calculus(II) (Spring 2020), Calculus(I) (Fall 2019), General Calculus (Spring 2019, Fall 2018)
Member, *Institute of Mathematical Statistics (IMS)*, *American Mathematical Society (AMS)*
Reviewer, *Annals of Applied Probability*, *Stochastic Processes and Their Applications*